

MUSTAFACODE

Lecture 11

Chapter: const, Initializer List & Static Data Members (Deep Notes)
Object-Oriented Programming (CS304)

Course Code: **CS304**

Publisher: **mustafacode**

Compiled On: **May 29, 2026**

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◆ 1. const Data Member

const kya hota hai?

`const` ka matlab hota hai:

👉 "Ek dafa value set ho gayi to phir change nahi hogi."

Class mein const member

```
class Student{  
    const int rollNo;  
};
```

👉 Iska matlab:

- Har student ka `rollNo` fixed hoga
 - Baad mein change nahi ho sakta
-

Problem samajho

Agar hum try karein:

```
Student::Student(int aNo){  
    rollNo = aNo; // ❌ ERROR  
}
```

✖ Error kyun aata hai?

Kyunkay:

- const member ko assignment se value nahi de sakte
 - Woh sirf initialization accept karta hai
-

📌 Real Life Example

👉 Roll number = CNIC number jaisa hota hai

- Ek dafa assign ho gaya
 - Phir change nahi hota
-

◆ 2. Initialization vs Assignment

📌 Initialization

👉 Variable banatay waqt value dena

```
int i = 5;
```

📌 Assignment

👉 Variable banne ke baad value dena

```
int i;  
i = 5;
```

Difference yaad rakho

Type	Time	Example
Initialization	Creation time	<code>int i = 5</code>
Assignment	Later	<code>i = 5</code>

const problem ka root

👉 const sirf initialization accept karta hai, assignment nahi.

◆ 3. Member Initializer List

Problem

Constructor ke andar const ko assign nahi kar sakte.

Solution

👉 Initializer list use karo.

Syntax

```
ClassName(parameters) : member(value)
{
}
```

Example

```
#include <iostream>
using namespace std;

class Student{
    const int rollNo;
    float GPA;

public:
    Student(int aRollNo)
        : rollNo(aRollNo), GPA(0.0)
    {
        cout << "Roll No: " << rollNo << endl;
        cout << "GPA: " << GPA << endl;
    }
};

int main(){

    Student s1(101);

    return 0;
}
```

Output

```
Roll No: 101
GPA: 0
```

Iska matlab

Member	Value
rollNo	Constructor se aayi
GPA	Default 0

Real Life Example

 Admission system:

- rollNo → fixed
 - GPA → start mein 0
 - Name → baad mein update ho sakta hai
-

Multiple members initialize karna

```
: rollNo(a), GPA(0), name("Unknown")
```

◆ 4. Important Rule — Initialization Order

Rule

 Members hamesha class ke order mein initialize hote hain

 Initializer list ke order mein nahi

Example

```
#include <iostream>
using namespace std;

class ABC{

    int x;
    int y;
    int z;

public:

    ABC() : y(10), x(y), z(y)
    {
        cout << "x = " << x << endl;
        cout << "y = " << y << endl;
        cout << "z = " << z << endl;
    }
};

int main(){

    ABC obj;

    return 0;
}
```

Actual Initialization Order

1. x
2. y
3. z

Problem

```
x(y)
```

👉 y abhi initialize nahi hua tha

Is liye:

- x = garbage value
- y = 10
- z = 10

Golden Rule

👉 "Order matters in class, not in initializer list."

◆ 5. const Object

const object kya hota hai?

Aisa object jiska data change nahi ho sakta.

```
const Student s1;
```

Rule

- ✓ Object read-only hota hai
 - ✗ Data modify nahi kar sakte
-

Example Problem

```
const Student s1;  
  
s1.getRollNo(); // ✗ Error
```

Error kyun aata hai?

Kyunkay normal function object ko change kar sakta hai.

Compiler safe rehna chahta hai.

◆ 6. const Member Function

Definition

Aisa function jo object ko modify na kare.

Syntax

```
int getRollNo() const;
```

Correct Example

```
#include <iostream>
using namespace std;

class Student{

    int rollNo;

public:

    Student(int r){
        rollNo = r;
    }

    int getRollNo() const{
        return rollNo;
    }
};

int main(){

    const Student s1(101);

    cout << s1.getRollNo();

    return 0;
}
```

Output

101

Real Life Example

👉 "Sirf dekh sakte ho, change nahi kar sakte."

Rule

✓ const object → sirf const function call kar sakta hai

◆ 7. Static Data Member

Definition

Aisa variable jo:

- Class ka hota hai
 - Object ka nahi hota
-

Simple Meaning

👉 Sab objects is variable ko share karte hain.

Example

```
class Student{  
  
    public:  
  
        static int count;  
  
};
```

Real Life Example

👉 University total students counter:

- s1 create → count++
- s2 create → count++
- s3 create → count++

👉 Sab same count share karte hain.

◆ 8. Static Member Memory Behavior

Important Points

- ✓ Sirf 1 copy banti hai
 - ✓ Sab objects use karte hain
 - ✓ Memory object ke andar nahi hoti
-

Static member define outside class

```
int Student::count = 0;
```

Complete Example

```
#include <iostream>
using namespace std;

class Student{

public:

    static int count;

    Student(){
        count++;
    }
};

int Student::count = 0;

int main(){

    Student s1;
    Student s2;
    Student s3;

    cout << "Total Students: " << Student::count;

    return 0;
}
```

Output

```
Total Students: 3
```

◆ 9. Important Interview / Exam Points

const Data Member

✓ Must initialize using initializer list

const Function

- ✓ Cannot modify data members
-

const Object

- ✓ Can call only const member functions
-

Static Data Member

- ✓ Shared by all objects
 - ✓ Single memory location
 - ✓ Defined outside class
-

◆ **FINAL SUMMARY**

One Line Revision

- `const` → value change nahi hoti
 - `initializer list` → constructor se pehle initialization
 - `const object` → read-only object
 - `const function` → object modify nahi karta
 - `static` → shared data among all objects
-

◆ Quick Revision Table

Concept	Simple Meaning
const variable	Change nahi ho sakti
Initialization	Creation time value
Assignment	Baad mein value dena
Initializer List	Constructor se pehle initialize
const object	Read-only object
const function	Data modify nahi karta
static member	Sab objects share karte hain
